Description

Illustration





Figure caption

1, 2. AAC can be used in most applications where traditional concrete is used, such as paving or general construction. 3. AAC may require more well-controlled curing conditions than traditional OPC concrete, potentially meaning it is better suited to precast applications. Images used under license from https://stock.adobe.com/

The material

Alkali-activated concrete (AAC) is defined as a class of concrete that is made without using any Portland cement. In AAC, the ordinary Portland cement (OPC) is entirely replaced by an alternative binder. These alternative materials are often industrial by-products such as ground-granulated blast-furnace slag (GGBS), coal fly ash, steel ash, etc. in combination with an alkaline activator - the two commonly used alkaline activators are caustic alkali and alkaline salts. Alkaline activators are often blended with ground-granulated blast-furnace slag (GGBS) before mixing it with water. As the characteristics of the alternative binder vary from source to source, the hydration reaction and resulting microstructure of the materials can also vary.

Alkali-activated concrete has great promise in the area of sustainability due to its significantly (up to 40-80%) lower carbon footprint than traditional (OPC) concrete. By using industrial by-products it also diverts materials typically destined for landfill. However, it is not yet widely used due to a lack of research into its long-term performance. International design codes are based on predicting lifetime performance based on performance after e.g. 28 days, and for alkali-activated concrete it is unclear whether the same lifetime predictions of compressive strength from 28-day behaviour can be drawn. Theoretically, AAC could offer many potential benefits over traditional concrete as well as the lower carbon footprint: reported benefits include excellent heat resistance with higher strength, lower porosity and lower shrinkage than concrete made with ordinary Portland cements.

Compositional summary

A mix of aggregate, binder and alkali activator

General properties

Density	2.16e3	-	2.3e3	kg/m^3

Material form that data applies to

Bulk	✓

Building system

Superstructure	✓
Enclosure	✓
Interiors	✗
Services	✗

Mechanical properties

Young's modulus		10	-	35	GPa
Shear modulus	*	4.6	-	15.8	GPa
Bulk modulus	*	4.6	-	17.5	GPa
Bending modulus	*	14	-	35	GPa
Poisson's ratio		0.13	-	0.18	
Yield strength (elastic limit)	*	1.3	-	8.4	MPa
Tensile strength		1.3	-	8.4	MPa
Compressive strength	*	16.2	-	63	MPa
Bending strength		1.3	-	9	MPa
Elongation	*	0	-	0.01	% strain
Fatigue strength at 10^7 cycles		2.4	-	3	MPa
Fracture toughness		0.652	-	1.58	MPa.m^0.5
Mechanical loss coefficient (tan delta)		0.01	-	0.03	

Thermal and combustion properties

Thermal conductor or insulator?		Poor insulator			
Thermal resistivity		0.46	-	0.67	m.°C/W
Thermal expansion coefficient		7	-	9	µstrain/°C
Specific heat capacity		770	-	1.05e3	J/kg.°C
Melting point		930	-	1.2e3	°C
Maximum service temperature		480	-	510	°C
Minimum service temperature		-160	-	-150	°C
Flammability		Non-flammable			
Emissivity		0.63	-	0.97	

Hygro-thermal properties

Water vapor permeability		2e-12	-	6.01e-12	kg.m/s.m^2.Pa
Frost resistance		Average			

Electrical properties

Electrical conductor or insulator?	Poor insulator			
Dielectric constant (relative permittivity)	* 8	-	12	
Dissipation factor (dielectric loss tangent)	* 0.001	-	0.01	

Optical properties

Natural color

Gray

Transparency	Opaque	
Transmissivity	0	%

Critical Materials Risk

High critical material risk?	Yes

Acoustic properties

Sound absorption	Average
Sound isolation	Good

Processability (scale 1 = impractical to 5 = excellent)

Castability	4	-	5
Moldability	3	-	4

Durability

Water (fresh)	Excellent
Water (salt)	Acceptable
Weak acids	Acceptable
Strong acids	Unacceptable
Weak alkalis	Acceptable
Strong alkalis	Acceptable
Organic solvents	Excellent
UV radiation (sunlight)	Excellent
Wear resistance	Unacceptable
Industrial atmosphere	Limited use
Rural atmosphere	Acceptable
Marine atmosphere	Acceptable

Primary material production: energy and climate change

Climate change (CO_2-eq) (virgin grade production)	* 0.0613	-	0.0674	kg/kg
Embodied energy (virgin grade production)	* 0.752	-	0.832	MJ/kg

Material processing: energy

Machining, grinding, embodied energy	* 18	-	19.6	MJ/kg

Material processing: climate change

Machining, grinding, climate change (CO2-eq)	* 1.17	-	1.28	kg/kg

Recycling and end of life

Recycle	✓			
Climate change (CO2-eq), recycling	* 0.0631	-	0.0698	kg/kg
Embodied energy, recycling	* 0.758	-	0.838	MJ/kg
Recycle fraction in current supply	13	-	14.4	%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
A renewable resource?	✗			

Supporting information

Design guidelines

Concrete made using alkali-activated cement concrete (AACC), commonly mis-termed as "Geopolymer" concrete, has been developed over the last 20 years as a sustainable version of Portland cement concrete (see Cement and Concrete).

Using normal dense aggregates, AACC is made with nearly 100% ground-granulated blast-furnace slag (GGBS), which is activated by addition of a strong alkali solution.

AACC is similar to concrete made with standard CEM III/C cement, specified in EN 197-1, which can comprise up to 95% GGBS and 5% CEM I (Portland cement), and where the hydrating cement provides sufficient alkali to activate the GGBS.

Technical notes

The use of AACC is becoming increasingly the subject of research and demonstration projects, with the HS2 trainline in England considering the possibility of AACC.
Recently, the UK has published a Publicly Available Standard to assist specifiers and producers in using AACC.
Alternative powders to GGBS include pozzolans, municipal solid waste incinerator ash (MSWIA), metakaolin, steel slag or other slags, or other alumina-rich materials.

The alkali used as the activator tends to be an alkali silicate solution such as sodium silicate (waterglass) but can also be sodium hydroxide solution, or a combination of the two, or other source of alkali (such as lime). These caustic materials must be used with care and so manufacture is commonly in precasting works.

Typical uses

Reinforced and unreinforced concrete civil engineering and bridge projects.

Tradenames

Cemfree

Related standards

The properties of Cement and Concrete vary a great deal based on composition, age and processing. Standards exist to help ensure that consistent material properties can be achieved. As temperature ranges, humidity, raw materials and building practices vary from country to country so do standards. Here you can see a list of the main US, EU and UK standards related to this material. Other countries have their own standards, for example: Japan www.jsa.or.jp/en, China www.gbstandards.org, Australia www.standards.org.au.

ASTM C150: Standard Specification for Portland Cement
ASTM C1697: Standard Specification for Blended Supplementary Cementitious Materials
EN 197-1: Cement. Composition, specifications and conformity criteria for common cements
EN 206: Concrete. Specification, performance, production and conformity
BS 8500: Concrete. Complementary British Standard to BS EN 206
AS 3972: General purpose and blended cements
JSCE Guidelines for concrete: Standard Specifications for Concrete Structures 2007
KIT 206: Design of concrete structures

Links

Reference